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RESEARCH ARTICLE

Addressing class imbalance in bankruptcy prediction: a cost-aware and explainable machine learning approach

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ABSTRACT

Corporate bankruptcy prediction is important for financial risk management, credit assessment, and early warning systems, yet it remains challenging under extreme class imbalance and asymmetric misclassification costs. This study develops a leakage-free and decision-oriented machine learning framework for bankruptcy prediction that integrates feature de-redundancy, cost-sensitive learning, probability calibration, threshold optimization, and explainable prediction within a unified evaluation pipeline. The empirical analysis is conducted on the Taiwanese Bankruptcy Prediction benchmark using repeated stratified cross-validation, two correlation thresholds, and multiple cost configurations. Rather than relying on discrimination metrics alone, the study evaluates model performance through precision–recall behavior, recall-oriented measures, calibration quality, expected misclassification cost, threshold stability, and statistical comparison. The revised results show that the Random Forest configuration consistently outperforms the Logistic Regression baseline in decision-oriented evaluation. Across alternative cost ratios, model rankings remain stable, and the preferred configuration yields lower expected cost, stronger minority-class detection, and more stable threshold behavior. Calibration analysis further indicates that predicted probabilities remain compressed under severe imbalance, highlighting the importance of combining probability calibration with cost-sensitive threshold selection. Overall, the study contributes a more rigorous and practically relevant benchmark framework for bankruptcy prediction in settings where model outputs support financially consequential decisions.

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1. Introduction

Bankruptcy prediction remains an important topic in financial risk analysis because it supports credit assessment, early-warning monitoring, and firm-level risk screening. Early research relied on ratio-based statistical models, while later studies increasingly adopted machine learning methods to capture more complex relationships among financial indicators (Altman, 1968; Barboza et al., 2017; Dellepiane et al., 2015; Min & Lee, 2005). Despite these advances, bankruptcy prediction remains difficult in practice because distressed firms typically represent only a small fraction of the observed population, creating a severe class-imbalance problem (Sun et al., 2014; Nkiri & Ofoegbu, 2022).

Learning from imbalanced data has therefore become a central methodological issue in bankruptcy prediction. Data-level approaches, especially synthetic oversampling methods such as SMOTE, are widely used to rebalance minority-class representation (Chawla et al., 2002; Fernández et al., 2018a, 2018b). Subsequent studies have proposed refinements and hybrid resampling strategies, with mixed empirical results depending on the data structure and the classifier used (Bajer et al., 2019; Zhou et al., 2013). Additional work has explored feature-level and relabeling strategies to stabilize learning under imbalance (Bui et al., 2020; Dao et al., 2020). However, prior findings remain inconsistent, particularly with respect to whether such interventions improve probability reliability and decision-oriented performance.

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An alternative line of work addresses imbalance at the algorithmic level through class weighting, ensemble learning, and cost-sensitive formulations. Random Forest and related ensemble approaches have shown strong performance in structured financial prediction tasks because they can capture non-linear interactions and are less sensitive to local noise than simple linear baselines (Barboza et al., 2017; Breiman, 2001; Sipper & Moore, 2021). Cost-sensitive learning extends this logic by recognizing that the consequences of different error types are asymmetric: failing to identify a distressed firm is often more consequential than incorrectly flagging a healthy one (Peykani et al., 2025; Radovanovic & Haas, 2023; Sahin et al., 2013). Yet, many studies still rely on a single fixed cost ratio and do not examine whether the resulting conclusions remain stable under alternative decision assumptions.

A related issue concerns model evaluation. In highly imbalanced settings, conventional metrics such as overall accuracy may be misleading because they can mask poor minority-class detection. This has motivated the use of precision–recall analysis, balanced accuracy, and other imbalance-aware measures (Brodersen et al., 2010; Carrillo et al., 2014; Saito & Rehmsmeier, 2015). However, the combined effects of class imbalance, probability calibration, and threshold-based decision-making remain insufficiently understood. In particular, rare-event settings often produce compressed probability estimates, which may affect both calibration quality and threshold optimization.

More recently, explainable artificial intelligence (XAI) has become increasingly important in financial machine learning. Foundational work on interpretable machine learning emphasizes that transparent model behavior is essential when predictive outputs may affect high-stakes decisions (Doshi-Velez & Kim, 2017; Guidotti et al., 2019). Recent reviews further show that explainability is becoming a core requirement in financial AI, not only for interpretation, but also for governance, auditability, and institutional trust (Yeo et al., 2025). At the same time, explainability is still often treated as a post hoc supplement rather than as part of a broader decision-oriented evaluation design.

In addition to these issues, methodological validity remains a concern. Data leakage can inflate reported performance when preprocessing, feature selection, or resampling is performed before the training–test split is respected (Kaufman et al., 2012). Financial-ratio datasets also contain substantial redundancy, which can affect model stability and interpretability if left untreated. These concerns point to the need for evaluation frameworks that are simultaneously leakage-free, imbalance-aware, cost-sensitive, and transparent.

Taken together, the existing literature reveals three important gaps. First, imbalance handling, cost-sensitive learning, calibration, and explainability are often examined in isolation rather than within a single integrated evaluation framework. Second, the robustness of model rankings under alternative cost ratios and preprocessing settings remains insufficiently tested. Third, the interaction between severe class imbalance, calibrated probability estimates, and threshold-based decision rules is still underexplored in bankruptcy prediction.

To address these limitations, this study develops a leakage-free and decision-oriented machine learning framework for bankruptcy prediction under extreme class imbalance. The framework integrates feature de-redundancy, class-imbalance handling, cost-sensitive learning, calibration analysis, threshold optimization, and explainable prediction within a unified experimental pipeline. Rather than claiming novelty in the form of a new standalone prediction algorithm, the contribution of the study lies in the joint integration and rigorous evaluation of these components under a controlled benchmark setting.

This positioning is important for two reasons. First, the individual methodological components used in the framework are already established in the literature. The novelty of the present study, therefore, lies in how they are combined and evaluated together in a leakage-free, decision-oriented design. Second, the Taiwanese Bankruptcy Prediction dataset used in this study is a widely adopted benchmark, but it reflects a specific institutional and temporal context. Accordingly, the findings should be interpreted primarily as evidence of methodological robustness within a benchmark setting rather than as definitive evidence of universal generalizability across all contemporary bankruptcy prediction environments. This more careful positioning is consistent with recent calls for stronger methodological transparency and more appropriately bounded practical claims in AI-enabled business and management research (Dvouletý et al., 2025).

To further clarify the empirical objectives of the study, we formulate the following research questions and hypotheses.

RQ1. How effective is cost-sensitive learning in improving decision performance under extreme class imbalance? Extreme class imbalance biases models toward the majority class, whereas cost-sensitive learning aims to mitigate this problem by penalizing false negatives more heavily than false positives (He & Garcia, 2009; Peykani et al., 2025; Sahin et al., 2013).

H1. Cost-sensitive learning reduces expected cost and improves recall-oriented performance relative to cost-insensitive evaluation.

RQ2. How robust are model performance and decision outcomes under different cost configurations? Most prior studies adopt fixed cost ratios without systematically testing the sensitivity of model rankings to cost variation (Radovanovic & Haas, 2023).

H2. Model rankings remain substantively stable across multiple misclassification cost configurations.

RQ3. How does extreme class imbalance affect probability estimation and threshold-based decision-making? Under rare-event conditions, probability estimates may be compressed and may therefore influence threshold optimization and cost-sensitive decisions (Saito & Rehmsmeier, 2015).

H3. Extreme class imbalance produces compressed calibrated probabilities, which in turn affects threshold selection and expected cost.

RQ4. Do ensemble-based models provide more reliable and robust performance than linear models in imbalanced bankruptcy prediction? Ensemble methods are often better able than linear models to represent interaction effects in structured financial data (Barboza et al., 2017; Breiman, 2001).

H4. Ensemble-based models outperform linear baselines in predictive and decision-oriented evaluation under extreme class imbalance.

Based on these gaps, the main contributions of the study are summarized as follows.

C1. A leakage-free and decision-oriented evaluation framework. The study develops a unified experimental pipeline for bankruptcy prediction that integrates feature de-redundancy, imbalance-aware learning, calibration, and threshold-based decision evaluation while preventing data leakage through training-fold-only operations (Kaufman et al., 2012).

C2. Robustness analysis across multiple decision settings. Rather than relying on a single cost ratio or a single correlation threshold, the revised framework evaluates model behavior across multiple cost configurations and alternative preprocessing settings, thereby allowing the stability of conclusions to be assessed under different decision assumptions.

C3. Joint analysis of class imbalance, calibration, and threshold behavior. The study provides empirical evidence on how severe class imbalance affects calibrated probability estimates and how these estimates interact with threshold optimization and expected cost.

C4. A transparent benchmark framework for financially relevant model comparison. By combining decision-oriented metrics, statistical comparison, calibration diagnostics, and explainable analysis, the framework provides a more comprehensive basis for comparing bankruptcy prediction models in settings where predictive outputs may inform financially consequential decisions.

In this sense, the contribution of the study lies not in proposing a new prediction algorithm, but in providing a more rigorous and practically relevant benchmark framework for bankruptcy prediction under extreme class imbalance.

The remainder of this paper is organized as follows. Section 2 reviews related work on bankruptcy prediction and imbalanced learning. Section 3 presents the proposed methodology and modeling framework. Section 4 reports the empirical results. Section 5 discusses the findings and their managerial, policy, and ethical implications. Finally, Section 6 concludes the study and outlines directions for future research.

2. Related work

2.1. Bankruptcy prediction from classical models to machine learning

Bankruptcy prediction has evolved from early ratio-based statistical approaches to more flexible machine learning models. Foundational studies established the role of financial ratios in identifying distressed firms and provided the basis for later predictive modeling in financial risk analysis (Altman, 1968; Sun

et al., 2014). Subsequent work introduced machine learning methods capable of capturing non-linear relationships and interactions among financial indicators, including support vector machines, decision trees, and ensemble-based approaches (Barboza et al., 2017; Dellepiane et al., 2015; Min & Lee, 2005).

Among machine learning methods, ensemble learners have attracted particular attention in structured financial prediction because they can model heterogeneous patterns in financial data while remaining relatively robust to local noise and variable interactions (Barboza et al., 2017; Breiman, 2001; Sipper & Moore, 2021). This shift from classical linear models toward more flexible predictive frameworks reflects a broader movement in bankruptcy and business failure prediction toward methods that are better suited to high-dimensional and non-linear financial settings.

At the same time, the increased flexibility of machine learning models has not eliminated key methodological concerns. In financial applications, predictive performance must be interpreted alongside questions of stability, transparency, evaluation validity, and practical usefulness in decision-making contexts. As a result, recent work increasingly emphasizes that model comparison should extend beyond raw classification performance and consider the broader decision environment in which predictions are used.

2.2. Learning under extreme class imbalance

A persistent difficulty in bankruptcy prediction is extreme class imbalance. In most bankruptcy datasets, distressed firms represent only a small minority of the observations, which causes conventional classifiers to favor the majority class and weakens minority-class detection (Fernández et al., 2018b; He & Garcia, 2009).

One common response is data-level imbalance handling, particularly through synthetic oversampling methods such as SMOTE and related variants (Chawla et al., 2002; Fernández et al., 2018b). These approaches aim to improve the representation of the minority class and can enhance recall in some settings. Hybrid or variant-based strategies have also been proposed and evaluated empirically, with mixed results depending on the classifier and the data structure (Bajer et al., 2019; Zhou et al., 2013). Additional work has explored relabeling and feature-level approaches for imbalanced learning, indicating that the imbalance problem is not limited to sampling alone but also interacts with representation and feature selection (Bui et al., 2020; Dang, 2024; Dang & Tuoi, 2024; Dao et al., 2020; Dao & Dang, 2023).

A second line of research addresses imbalance at the algorithmic level through class weighting and cost-sensitive learning. Rather than changing the data distribution, these methods alter the learning objective to assign greater importance to minority-class errors. This is particularly relevant in financially consequential settings where the cost of missing a distressed firm is usually higher than the cost of issuing a false alarm (Peykani et al., 2025; Radovanovic & Haas, 2023; Sahin et al., 2013). Existing evidence suggests that the benefits of these methods depend not only on the choice of model but also on the interaction between cost assumptions, threshold rules, and the evaluation criteria used.

Hybrid strategies that combine resampling and cost-sensitive learning have also been studied. However, their effectiveness remains model-dependent, and the literature does not support the assumption that resampling is universally beneficial in all imbalanced financial datasets (Fernández et al., 2018a, 2018b; Zhou et al., 2013). This motivates the need for leakage-free and decision-oriented comparison of multiple configurations rather than default reliance on a single imbalance-handling technique.

2.3. Evaluation validity, cost-sensitive assessment, calibration, and explainability

The evaluation of bankruptcy prediction models under class imbalance requires more than conventional accuracy-based metrics. In rare-event settings, accuracy and even ROC-AUC can present an incomplete picture because they may remain high despite weak performance on the minority class. For this reason, imbalance-aware metrics such as recall, F-measure variants, balanced accuracy, Matthews correlation coefficient, and precision–recall area under the curve (PR-AUC) have been widely recommended (Brodersen et al., 2010; Carrillo et al., 2014; Pakravan & Jahed, 2022; Saito & Rehmsmeier, 2015).

Recent literature has gone further by arguing that model evaluation should reflect the asymmetric consequences of misclassification. In bankruptcy prediction, missed distress signals may have greater financial and supervisory consequences than false alarms. This has led to growing interest in cost-sensitive learning, expected-cost evaluation, and threshold optimization as more realistic decision-oriented

paradigms (Peykani et al., 2025; Radovanovic & Haas, 2023; Sahin et al., 2013). This shift is consistent with broader developments in AI-enabled business and management research, where methodological transparency, decision relevance, and practical robustness are increasingly emphasized (Dvouletý et al., 2025).

Another methodological issue is data leakage. Leakage occurs when information from test data influences model construction through preprocessing, feature selection, scaling, or resampling performed outside the training split. Prior work has shown that leakage can substantially inflate reported performance and undermine reproducibility, especially in small or imbalanced datasets (Kaufman et al., 2012). In applied financial prediction, this makes leakage-free evaluation an essential condition for credible empirical comparison.

Calibration provides a further dimension of model validity. In decision-oriented settings, predicted probabilities are not only used for ranking, but may also support threshold-based actions, escalation, or screening decisions. Under severe class imbalance, however, probability estimates may become compressed or poorly calibrated, complicating the interpretation of risk scores. While calibration is widely recognized as important, the interaction between class imbalance, calibrated probabilities, and threshold optimization remains underexamined in bankruptcy prediction.

Finally, explainability has become increasingly important in financial machine learning. Foundational work in interpretable machine learning and later surveys emphasize the need for model outputs that can be understood, scrutinized, and related to domain-relevant variables (Doshi-Velez & Kim, 2017; Guidotti et al., 2019). In financial AI, explainability also supports governance, auditability, and institutional trust (Yeo et al., 2025). However, explainability is still often treated as a post hoc add-on rather than as part of an integrated evaluation framework that also considers cost sensitivity, calibration, and threshold-dependent decision rules.

2.4. Research gap and positioning of the study

Taken together, the recent literature points toward a more integrated research agenda in which bankruptcy prediction models are expected to be not only discriminative, but also cost-sensitive, transparent, and methodologically robust. Nevertheless, these dimensions are still frequently studied in isolation. Some studies focus primarily on model class, some on resampling, some on asymmetric costs, and others on explainability, with limited attention to how these components interact within a single evaluation design.

The present study is positioned within this literature as a decision-oriented benchmark framework rather than as a proposal for a new standalone prediction algorithm. Its contribution lies in evaluating how feature de-redundancy, leakage control, cost-sensitive learning, calibrated probability assessment, threshold optimization, and explainable AI operate jointly under extreme class imbalance. In addition, the revised experiments extend this positioning by testing the robustness of conclusions across multiple cost ratios and correlation thresholds.

To clarify these differences, Table 1 provides a structured comparison of representative prior studies and the present framework.

As shown in Table 1, the originality of the present study does not derive from any single methodological component considered in isolation. Rather, it lies in the way these components are combined and evaluated together within a leakage-free, cost-sensitive, calibration-aware, and explainability-supported

Table 1. Structured comparison of prior studies and the proposed framework.

Study	Imbalance handling	Cost-sensitive learning	Leakage control	Calibration analysis	Explainability	Sensitivity across costs
Altman (1968)	X	X	X	X	X	X
Barboza et al. (2017)	Limited	X	X	X	X	X
Zhou et al. (2013)	✓ (sampling)	X	X	X	X	X
Fernández et al. (2018b)	✓ (SMOTE variants)	X	X	X	X	X
Sahin et al. (2013)	Partial	✓	X	X	X	X
Peykani et al. (2025)	Partial	✓	X	X	X	X
Radovanovic and Haas (2023)	Partial	✓	X	X	X	X
Guidotti et al. (2019)	X	X	X	X	✓	X
Yeo et al. (2025)	X	X	X	X	✓	X
This study	✓	✓	✓	✓	✓	✓

framework. The revised experiments further extend this contribution by testing whether the resulting conclusions remain stable across multiple cost scenarios and correlation thresholds.

Accordingly, the present study contributes not a new standalone algorithm, but a more rigorous and practically relevant benchmark framework for bankruptcy prediction under extreme class imbalance. This positioning is particularly important in financially consequential settings, where model performance should be judged not only by average predictive accuracy, but also by robustness, calibration quality, transparency, and the stability of decision-relevant conclusions.

3. Methods

This study develops a leakage-free and decision-oriented machine learning framework for corporate bankruptcy prediction under extreme class imbalance. The framework integrates feature de-redundancy, imbalance-aware learning, cost-sensitive decision rules, and explainable analysis within a unified evaluation pipeline. To ensure methodological rigor, all data-dependent operations are performed exclusively within the training folds of cross-validation, thereby preventing information leakage and supporting reliable performance assessment in an imbalanced setting.

Figure 1 provides an overview of the proposed framework. The pipeline consists of four main stages: (1) leakage-free data partitioning and preprocessing, (2) feature de-redundancy, (3) imbalance-aware and cost-sensitive learning, and (4) explainable prediction. These components are implemented within a unified pipeline to ensure consistency across all experimental settings.

3.1. Dataset

The empirical analysis uses the Taiwanese Bankruptcy Prediction dataset, which is publicly available from the UCI Machine Learning Repository (Taiwanese Bankruptcy Prediction, 2020). The dataset contains firm-level financial information collected from the Taiwan Economic Journal over the period 1999–2009. Each observation corresponds to a firm-year instance described by 95 financial ratios, and the binary target variable indicates whether the firm experienced bankruptcy-related events.

From a managerial perspective, these variables summarize core dimensions of firm condition, including profitability, liquidity, leverage, solvency, operational efficiency, and growth. These dimensions are commonly monitored in financial analysis and early-warning assessment. A defining characteristic of the dataset is its extreme class imbalance, with bankrupt firms representing a small minority of the observations. This imbalance motivates the use of imbalance-aware and cost-sensitive learning strategies.

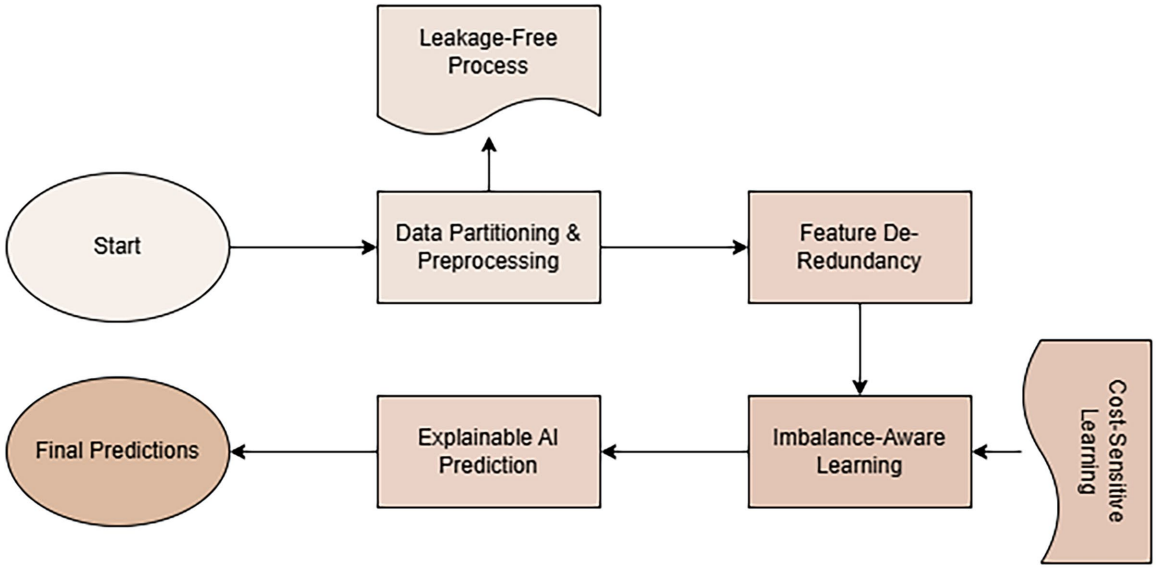


Figure 1. Overview of the proposed AI framework for bankruptcy prediction.

The dataset is used here as a widely adopted benchmark for bankruptcy prediction under class imbalance. However, because it represents a single institutional setting and time period, the empirical findings should be interpreted within the scope of this benchmark rather than as direct evidence of universal generalizability across all financial environments.

3.2. Leakage-free evaluation protocol

To ensure unbiased performance estimation, the experimental design follows a strictly leakage-free evaluation protocol based on K-fold cross-validation.

Let the dataset be defined as:

$$D = \{(x_i, y_i)\}_{i=1}^{N},$$

where x_i denotes the feature vector and $y_i \in \{0,1\}$ represents the bankruptcy label.

For each fold k, the data are partitioned into training and test subsets:

$$D = D_{\text{train}}^{(k)} \cup D_{\text{test}}^{(k)}, D_{\text{train}}^{(k)} \cap D_{\text{test}}^{(k)} = \varnothing.$$

All data-dependent operations, including preprocessing, feature selection, resampling, and model training, are performed exclusively on $D_{\text{train}}^{(k)}$. The learned transformations are then applied to $D_{\text{test}}^{(k)}$.

This protocol ensures that no information from the test data is used during model construction.

3.3. Feature de-redundancy

Financial ratios often exhibit strong linear dependencies, which may introduce multicollinearity and reduce model stability. To address this issue, a two-stage feature de-redundancy procedure is applied within each training fold.

3.3.1. Correlation-based filtering

For any pair of features x_j and x_k, the Pearson correlation coefficient is computed as:

$$\rho_{jk} = \frac{\text{Cov}(x_j, x_k)}{\sigma_{x_j}\sigma_{x_k}}.$$

If $|\rho_{jk}| > \tau$, where τ is a predefined threshold, one of the features is removed based on its relevance to the target variable.

In this study, τ is treated as a conservative screening parameter used to identify highly redundant feature pairs. Rather than assuming a single universally optimal value, the revised analysis evaluates two alternative thresholds,

$$\tau \in \{0.90, 0.95\},$$

which represent two closely related levels of strictness for high-correlation filtering. All subsequent stages of the pipeline are re-estimated under both settings so that the impact of τ on predictive and decision-oriented performance can be assessed empirically.

3.3.2. Model-based feature importance

Tree-based models are used to estimate feature importance scores within each training fold. The importance values are aggregated across folds:

$$I_j = \frac{1}{K}\sum_{k=1}^{K} I_j^{(k)}.$$

Features with consistently high importance scores are retained for model training. All feature-selection steps are performed within the training data only.

3.4. Imbalance-aware and cost-sensitive learning

To address extreme class imbalance and asymmetric misclassification costs, model training is formulated as a cost-sensitive optimization problem. The learning objective is defined as:

$$L = \sum_{i=1}^{N} w_{y_i} \, l\left(y_i, \hat{y}_i\right),$$

where w_{y_i} denotes the class-specific weight, with $w_1 > w_0$ assigning a higher penalty to misclassifying bankrupt firms, and $l(\cdot)$ represents the base loss function. In plain terms, this formulation means that the model penalizes missed bankruptcies more heavily than false alarms, thereby placing greater emphasis on detecting financially distressed firms during training.

This setup allows the model to emphasize minority-class cases without changing the original data distribution. Optional resampling techniques, such as SMOTE-based methods, may also be applied within each training fold to improve minority-class representation. Resampling is performed after feature processing and is restricted to $D_{\text{train}}^{(k)}$.

3.5. Cost-sensitive decision threshold optimization

Let

$$\hat{p}_i = P(y_i = 1 | \, x_i)$$

denote the predicted probability of bankruptcy for observation i.

Instead of using a fixed threshold, the classification decision is determined by minimizing the expected misclassification cost:

$$t^* = \arg\min_t \left[C_{FN} \cdot FN(t) + C_{FP} \cdot FP(t) \right],$$

where $FN(t)$ and $FP(t)$ denote the numbers of false negatives and false positives at threshold t, and C_{FN} and C_{FP} represent the corresponding costs.

Because the dataset does not contain institution-specific exposure, collateral, recovery, or realized loss information, the cost ratios used in this study are interpreted as stylized decision parameters rather than direct monetary loss estimates. To avoid dependence on a single specification, the revised analysis evaluates four cost configurations:

$$\left(C_{FP}, C_{FN}\right) \in \left\{(1,1), (1,5), (1,10), (1,20)\right\}.$$

The optimal threshold t^* is estimated using $D_{\text{train}}^{(k)}$ and then applied to $D_{\text{test}}^{(k)}$.

3.6. Predictive models

Two representative classification models are employed:

- Logistic Regression (LR) as a linear baseline model;
- Random Forest (RF) as a bagging-based ensemble model.

All models are trained under the same leakage-free and cost-sensitive framework to ensure comparability across experimental settings.

3.7. Evaluation metrics

Model performance is evaluated using metrics suitable for imbalanced classification and decision-oriented analysis.

Discrimination Metrics

- Precision–Recall Area Under the Curve (PR-AUC)
- Receiver Operating Characteristic Area Under the Curve (ROC-AUC)

Recall-Oriented Metrics

- Recall
- F2-score
- Balanced Accuracy

Overall Classification Quality

- Matthews Correlation Coefficient (MCC)

Calibration and Cost-Based Metrics

- Brier Score
- Expected misclassification cost:

$$\text{Cost} = C_{FN} \cdot FN + C_{FP} \cdot FP$$

All metrics are computed on $D_{\text{test}}^{(k)}$ and aggregated across folds.

3.8. Statistical validation

To assess the statistical significance of performance differences, non-parametric tests are conducted on fold-level results. Pairwise model comparisons are performed using the Wilcoxon signed-rank test at a significance level of $\alpha = 0.05$. All tests are applied to paired results obtained from identical cross-validation folds.

In addition to Wilcoxon testing, effect sizes are quantified using Cliff's delta in order to assess the magnitude of pairwise differences at the fold level. This complements p-values by distinguishing statistically detectable effects from practically meaningful ones.

To further support reproducibility transparency, the implementation details of the revised benchmark analysis are documented through the reported experimental settings, and the data-processing scripts, analysis code, and derived outputs supporting the findings can be made available from the corresponding author upon reasonable request.

4. Results

4.1. Experimental setting

The revised empirical evaluation was conducted on the Taiwanese Bankruptcy Prediction dataset, comprising 6,819 observations, including 220 bankrupt and 6,599 non-bankrupt firms (imbalance ratio ≈ 1:30). The benchmark design used repeated stratified cross-validation with 5 folds and 2 repeats. All preprocessing, feature filtering, optional resampling, model fitting, calibration, and threshold optimization procedures were performed strictly within the training folds.

The final revised benchmark included two classifier families: LR_balanced and RF_balanced. For Logistic Regression, the configuration used the liblinear solver, max_iter = 2000, class_weight = balanced, and random_state = 777. For Random Forest, the configuration used n_estimators = 500, class_weight = balanced_subsample, n_jobs = 1, and random_state = 777. The evaluated preprocessing settings were none

and smoteenn, and probability calibration was performed using the sigmoid method. Sensitivity analysis was conducted under four asymmetric cost configurations,

$$(C_{FP}, C_{FN}) \in \{(1,1),(1,5),(1,10),(1,20)\},$$

and under two correlation thresholds,

$$\tau \in \{0.90, 0.95\}.$$

4.2. Predictive performance under the primary cost configuration

Table 2 reports the predictive performance of all evaluated model configurations under the primary cost configuration $C_{FP} : C_{FN} = 1:10$, with results presented separately for the two correlation thresholds.

The standard deviations in Table 2 complement the corresponding means by showing fold-level dispersion across repeated cross-validation.

4.3. Sensitivity of expected cost to cost ratio

Table 3 summarizes the expected test cost across the four cost configurations. The values are aggregated over the evaluated correlation thresholds.

Figure 2 presents the same relationship graphically.

Expected cost as a function of C_{FN} for the four evaluated model configurations, averaged over repeated cross-validation.

4.4. Sensitivity of decision thresholds to cost ratio

The optimized decision thresholds corresponding to each cost configuration are reported in Table 4. These values are also aggregated over the two evaluated correlation thresholds.

4.5. Sensitivity of recall and F2-score to cost ratio

Table 5 reports recall and F2-score across the four cost configurations.

Across Tables 3–5, expected cost increases as the false-negative penalty becomes larger, while recall and F2-score vary according to the cost structure and model configuration. The corresponding model ordering is reported consistently across all four evaluated cost scenarios.

4.6. Calibration results

Calibration statistics under the primary cost ratio 1:10 are reported in Table 6. Reliability diagrams for the four model–resampler combinations are shown in Figure 3.

Calibration performance is evaluated using both numerical and graphical diagnostics. Table 6 reports the Brier score and expected calibration error (ECE), while Figure 3 shows the relationship between predicted probabilities and observed frequencies across probability bins for each model–resampler configuration.

4.7. Pairwise statistical comparison

Pairwise statistical comparisons were conducted using the Wilcoxon signed-rank test. Cliff's delta was used to quantify effect size. Table 7 reports selected pairwise comparisons for PR-AUC, expected test cost, and Brier score.

Detailed analytical interpretation of these empirical patterns is provided in the Discussion section in order to maintain a clear separation between reporting of results and theoretical implications.

Table 2. Predictive performance under the 1:10 cost ratio (mean ± standard deviation across folds).

Model	τ	PR-AUC	ROC-AUC	Recall	F2-score	Precision	MCC	Balanced accuracy	Brier	ECE	Expected cost	Threshold
RF_balanced__none	0.90	0.4083 ± 0.0870	0.9386 ± 0.0204	0.7386 ± 0.0898	0.5421 ± 0.0516	0.2675 ± 0.0391	0.4145 ± 0.0458	0.8347 ± 0.0414	0.0253 ± 0.0024	0.0096 ± 0.0027	206.3000 ± 34.0557	0.0388 ± 0.0076
RF_balanced__smoteenn	0.90	0.3829 ± 0.0909	0.9388 ± 0.0201	0.7841 ± 0.0942	0.5192 ± 0.0478	0.2236 ± 0.0261	0.3855 ± 0.0402	0.8460 ± 0.0434	0.0247 ± 0.0021	0.0091 ± 0.0021	216.5000 ± 35.2302	0.0534 ± 0.0139
LR_balanced__smoteenn	0.90	0.3026 ± 0.1191	0.8112 ± 0.2602	0.4000 ± 0.3034	0.3271 ± 0.2388	0.2063 ± 0.1539	0.2630 ± 0.1918	0.6827 ± 0.1387	0.0293 ± 0.0016	0.0052 ± 0.0051	309.6000 ± 100.0169	0.0449 ± 0.0086
LR_balanced__none	0.90	0.3078 ± 0.1237	0.8141 ± 0.2601	0.4023 ± 0.3207	0.3161 ± 0.2298	0.2110 ± 0.1659	0.2599 ± 0.1872	0.6816 ± 0.1420	0.0293 ± 0.0019	0.0062 ± 0.0068	314.6000 ± 95.2998	0.0766 ± 0.0958
RF_balanced__none	0.95	0.4022 ± 0.0872	0.9382 ± 0.0207	0.7727 ± 0.1098	0.5306 ± 0.0459	0.2397 ± 0.0232	0.3984 ± 0.0367	0.8448 ± 0.0482	0.0253 ± 0.0024	0.0098 ± 0.0024	209.6000 ± 33.7843	0.0328 ± 0.0041
RF_balanced__smoteenn	0.95	0.3853 ± 0.0893	0.9386 ± 0.0208	0.7455 ± 0.1202	0.5173 ± 0.0627	0.2383 ± 0.0353	0.3884 ± 0.0515	0.8320 ± 0.0543	0.0247 ± 0.0020	0.0087 ± 0.0028	219.6000 ± 42.0428	0.0655 ± 0.0185
LR_balanced__smoteenn	0.95	0.3258 ± 0.0808	0.8927 ± 0.0399	0.3977 ± 0.3012	0.3286 ± 0.2399	0.2108 ± 0.1567	0.2655 ± 0.1927	0.6823 ± 0.1384	0.0291 ± 0.0018	0.0051 ± 0.0048	308.7000 ± 100.9753	0.0454 ± 0.0088
LR_balanced__none	0.95	0.3058 ± 0.1199	0.8459 ± 0.1543	0.3818 ± 0.3039	0.3177 ± 0.2370	0.2265 ± 0.1739	0.2653 ± 0.1912	0.6755 ± 0.1387	0.0294 ± 0.0018	0.0058 ± 0.0064	312.8000 ± 99.1999	0.0796 ± 0.0912

Table 3. Expected test cost across cost ratios (mean ± standard deviation).

Model	1:1	1:5	1:10	1:20
LR_balanced__none	44.20 ± 2.31	178.30 ± 34.56	313.70 ± 94.68	545.45 ± 236.29
LR_balanced__smoteenn	44.20 ± 1.70	180.40 ± 30.84	309.15 ± 97.82	532.30 ± 250.54
RF_balanced__none	43.45 ± 4.32	147.65 ± 18.47	207.95 ± 33.06	300.50 ± 57.10
RF_balanced__smoteenn	44.70 ± 5.16	148.65 ± 20.54	218.05 ± 37.79	306.95 ± 76.77

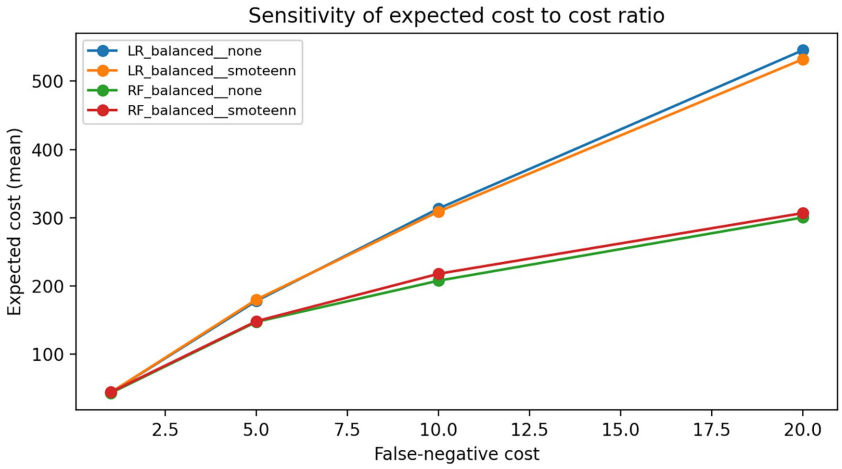


Figure 2. Sensitivity of expected cost to the false-negative cost.

Table 4. Threshold stability across cost ratios (mean ± standard deviation).

Model	1:1	1:5	1:10	1:20
LR_balanced__none	0.2049 ± 0.1289	0.0895 ± 0.0876	0.0781 ± 0.0911	0.0723 ± 0.0930
LR_balanced__smoteenn	0.2288 ± 0.1470	0.0542 ± 0.0148	0.0451 ± 0.0085	0.0386 ± 0.0026
RF_balanced__none	0.5320 ± 0.2083	0.0965 ± 0.0499	0.0358 ± 0.0067	0.0272 ± 0.0035
RF_balanced__smoteenn	0.4690 ± 0.0715	0.1782 ± 0.0601	0.0595 ± 0.0171	0.0406 ± 0.0198

Table 5. Recall and F2-score across cost ratios (mean ± standard deviation).

Cost ratio	Model	Recall	F2-score
1:1	LR_balanced__none	0.0966 ± 0.1198	0.1089 ± 0.1313
1:1	LR_balanced__smoteenn	0.0670 ± 0.0615	0.0789 ± 0.0714
1:1	RF_balanced__none	0.1682 ± 0.1087	0.1897 ± 0.1140
1:1	RF_balanced__smoteenn	0.1636 ± 0.0988	0.1834 ± 0.1035
1:5	LR_balanced__none	0.3080 ± 0.2693	0.2792 ± 0.2227
1:5	LR_balanced__smoteenn	0.3193 ± 0.2468	0.2872 ± 0.2083
1:5	RF_balanced__none	0.5318 ± 0.1036	0.4773 ± 0.0646
1:5	RF_balanced__smoteenn	0.5489 ± 0.1234	0.4805 ± 0.0763
1:10	LR_balanced__none	0.3920 ± 0.3043	0.3169 ± 0.2272
1:10	LR_balanced__smoteenn	0.3989 ± 0.2943	0.3279 ± 0.2330
1:10	RF_balanced__none	0.7557 ± 0.0992	0.5364 ± 0.0479
1:10	RF_balanced__smoteenn	0.7648 ± 0.1069	0.5183 ± 0.0543
1:20	LR_balanced__none	0.4773 ± 0.3379	0.3237 ± 0.2252
1:20	LR_balanced__smoteenn	0.4943 ± 0.3564	0.3257 ± 0.2257
1:20	RF_balanced__none	0.8148 ± 0.0883	0.5137 ± 0.0330
1:20	RF_balanced__smoteenn	0.8284 ± 0.1155	0.4979 ± 0.0565

Table 6. Calibration statistics under the 1:10 cost ratio (mean ± standard deviation across folds).

Model	τ	Brier score	ECE
RF_balanced__none	0.90	0.0253 ± 0.0024	0.0096 ± 0.0027
RF_balanced__smoteenn	0.90	0.0247 ± 0.0021	0.0091 ± 0.0021
LR_balanced__smoteenn	0.90	0.0293 ± 0.0016	0.0052 ± 0.0051
LR_balanced__none	0.90	0.0293 ± 0.0019	0.0062 ± 0.0068
RF_balanced__none	0.95	0.0253 ± 0.0024	0.0098 ± 0.0024
RF_balanced__smoteenn	0.95	0.0247 ± 0.0020	0.0087 ± 0.0028
LR_balanced__smoteenn	0.95	0.0291 ± 0.0018	0.0051 ± 0.0048
LR_balanced__none	0.95	0.0294 ± 0.0018	0.0058 ± 0.0064

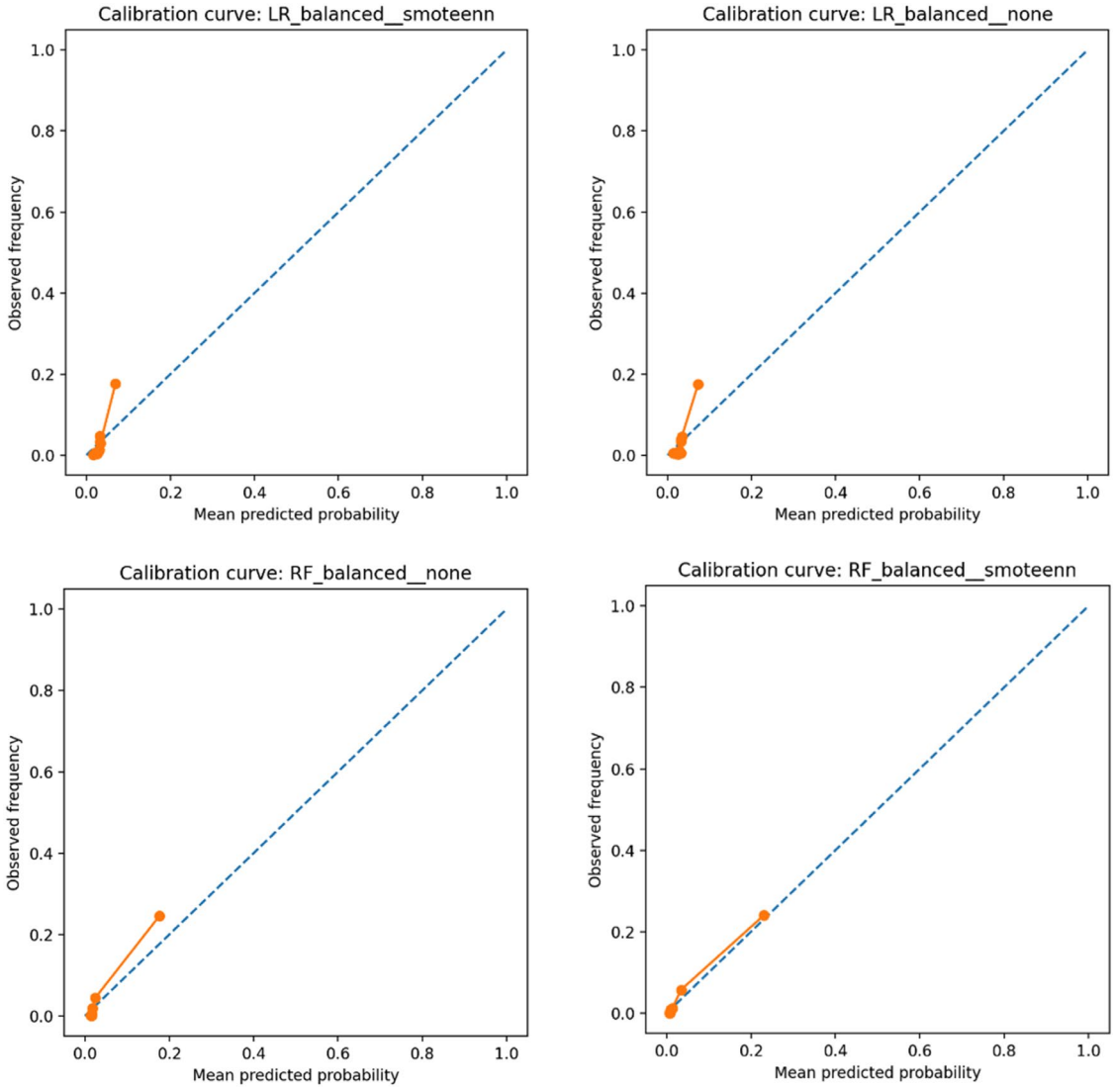


Figure 3. Reliability diagrams for calibrated probability estimates. Panels (a–d) present calibration curves for LR_balanced__none, LR_balanced__smoteenn, RF_balanced__none, and RF_balanced__smoteenn, respectively. The dashed diagonal line indicates perfect calibration.

Table 7. Selected pairwise statistical comparisons.

Comparison	Metric	Wilcoxon statistic	*p*-value	Cliff's Delta	Magnitude	Mean (model 1)	Mean (model 2)
LR_balanced__none vs LR_balanced__smoteenn	PR-AUC	1600.0	9.2356e-01	−0.0100	Negligible	0.3068	0.3142
LR_balanced__none vs LR_balanced__smoteenn	Expected cost	633.5	9.6919e-01	0.0144	Negligible	270.4125	266.5125
LR_balanced__none vs LR_balanced__smoteenn	Brier	908.0	6.3563e-04	0.1150	Negligible	0.029356	0.029183
LR_balanced__none vs RF_balanced__none	PR-AUC	94.0	2.4569e-13	−0.5050	Large	0.3068	0.4053
LR_balanced__none vs RF_balanced__none	Expected cost	230.5	6.7792e-11	0.2284	Small	270.4125	174.8875
LR_balanced__none vs RF_balanced__none	Brier	36.0	2.9727e-14	0.7950	Large	0.029356	0.025277
LR_balanced__none vs RF_balanced__smoteenn	PR-AUC	258.0	6.3855e-11	−0.3800	Medium	0.3068	0.3841
LR_balanced__none vs RF_balanced__smoteenn	Expected cost	225.0	5.6464e-11	0.2244	Small	270.4125	179.5875
LR_balanced__none vs RF_balanced__smoteenn	Brier	0.0	7.7122e-15	0.8700	Large	0.029356	0.024704
RF_balanced__none vs RF_balanced__smoteenn	PR-AUC	318.0	4.1958e-10	0.1500	Small	0.4053	0.3841
RF_balanced__none vs RF_balanced__smoteenn	Expected cost	1080.0	2.1728e-02	−0.0316	Negligible	174.8875	179.5875
RF_balanced__none vs RF_balanced__smoteenn	Brier	396.0	4.2968e-09	0.1300	Negligible	0.025277	0.024704

5. Discussion

5.1. Model performance under extreme class imbalance

The revised results show a consistent separation between the linear baseline and the ensemble-based approach under extreme class imbalance. Across both evaluated correlation thresholds and under the

primary 1:10 cost ratio, the Random Forest configurations achieve more favorable performance than the Logistic Regression baseline in terms of minority-class detection, decision-oriented cost, and threshold stability. This pattern is consistent with prior evidence that tree-based ensemble methods are better able than linear models to capture non-linear relationships and interaction effects among financial ratios in bankruptcy prediction (Barboza et al., 2017; Breiman, 2001; Dellepiane et al., 2015).

This distinction is especially relevant because the benchmark dataset is heavily skewed toward non-bankrupt firms. Under such conditions, global discrimination alone does not adequately reflect practical usefulness, and metrics focused on the minority class become more informative (He & Garcia, 2009; Krawczyk, 2016; Saito & Rehmsmeier, 2015). The stronger recall-oriented performance of Random Forest therefore indicates not only better ranking ability, but also greater usefulness for identifying rare bankruptcy events in a decision-support setting.

From a managerial perspective, higher recall means that a larger proportion of financially distressed firms can be identified before failure events occur. In banking and credit-risk contexts, this reduces the probability that vulnerable firms remain undetected during screening or monitoring. For regulators and supervisory bodies, the same result improves the value of model-assisted early-warning systems by increasing the likelihood that firms requiring closer review or earlier intervention are flagged in time.

The results also remain stable across $\tau = 0.90$ and $\tau = 0.95$, suggesting that the main conclusions do not depend materially on the specific correlation threshold used in the feature de-redundancy stage. This is relevant because redundancy in financial ratios has long been recognized as a challenge in bankruptcy prediction, and the present findings indicate that moderate changes in correlation filtering do not alter the overall ranking of the evaluated model configurations (Pohlman & Hollinger, 1981).

5.2. Cost-sensitive learning and robustness across cost ratios

A central objective of the revised analysis was to examine whether the ranking of models remains stable when the cost of false negatives increases. This issue is particularly important in bankruptcy prediction because the cost of failing to identify a distressed firm is usually higher than the cost of issuing a false alarm (Peykani et al., 2025; Radovanovic & Haas, 2023; Sahin et al., 2013).

The sensitivity analysis shows that the expected cost rises monotonically for all model configurations as C_{FN} increases from 1 to 20. More importantly, the relative ranking of models remains substantively unchanged: RF_balanced__none consistently yields the lowest expected cost, followed by RF_balanced__smoteenn, while both Logistic Regression configurations remain less favorable. This strengthens the decision-oriented contribution of the study by showing that the preferred configuration is not an artifact of a single, arbitrarily chosen cost ratio.

The slope of the cost increase also differs across model classes. Expected cost rises more steeply for Logistic Regression than for Random Forest, implying that model choice becomes increasingly consequential as the penalty for missed bankruptcies grows. This result extends prior work on cost-sensitive learning by showing not only that asymmetric costs matter, but also that model rankings can remain robust across multiple plausible cost structures rather than under one stylized specification.

Variability across folds provides an additional perspective on model robustness. In this setting, a model with slightly better average performance but unstable thresholds or widely dispersed expected costs may be less attractive in practice than a model with comparable averages and more stable decision behavior. The revised experiments show that the Random Forest configurations combine lower expected cost with more stable thresholds, strengthening the practical interpretation of their advantage.

Taken together, these findings provide strong support for **H2**: model rankings remain stable across all evaluated cost configurations, and the advantage of Random Forest over Logistic Regression persists despite substantial changes in the false-negative penalty.

5.3. Probability compression, calibration, and threshold behavior

The calibration analysis provides additional insight into the interaction between class imbalance and decision-making. The reliability diagrams show that calibrated probabilities remain concentrated in a low-probability region for all model configurations. This pattern is consistent with the rarity of bankruptcy events in the dataset and indicates a form of probability compression under extreme imbalance.

This finding matters because threshold-based decisions depend directly on the distribution of predicted probabilities. In the revised experiments, the optimized thresholds decrease as the false-negative cost increases, as expected under a cost-sensitive decision rule. However, the threshold trajectories differ across models. Random Forest shows a stronger downward adjustment from relatively high thresholds under the 1:1 setting to much lower thresholds under the 1:10 and 1:20 settings, whereas Logistic Regression operates within a narrower low-probability regime. This indicates that threshold optimization interacts with different probability structures rather than simply rescaling similar outputs.

The calibration statistics reveal an additional nuance. Random Forest achieves better Brier scores than Logistic Regression, whereas Logistic Regression attains slightly smaller ECE values in some configurations. These observations are not contradictory. The Brier score reflects overall probabilistic accuracy, whereas ECE summarizes bin-level calibration gaps. In a highly imbalanced setting, a model may appear locally aligned in bins while still offering limited probability separation for decision-making. The revised calibration analysis, therefore, suggests that ECE should not be interpreted in isolation, especially when the probability mass is concentrated in a narrow region.

Taken together, these findings support **H3**: extreme class imbalance is associated with compressed probability estimates, and this compression is directly linked to threshold behavior and cost outcomes.

5.4. The role of resampling in the revised experiments

The revised experiments provide a more nuanced view of the role of SMOTEENN. Resampling does not produce a uniform benefit across model classes or evaluation criteria. For Random Forest, SMOTEENN yields slightly lower Brier score and, in some settings, slightly higher recall, but it is accompanied by lower PR-AUC and higher expected cost than the non-resampled configuration. For Logistic Regression, the differences between none and smoteenn are generally small, with only marginal changes in cost and PR-AUC.

The statistical comparison reinforces this interpretation. For Logistic Regression, the differences in PR-AUC and expected cost between the two resampling settings are negligible in practical magnitude. For Random Forest, the change in PR-AUC is statistically detectable but small in effect size, while the difference in expected cost remains negligible in magnitude. This distinction is important because it shows that statistical significance alone is not sufficient to establish practical superiority. In the present benchmark, the revised evidence does not support the view that hybrid resampling is uniformly beneficial once class balancing, calibration, and threshold optimization are already incorporated into the evaluation pipeline.

These findings are aligned with earlier studies showing that SMOTE-based methods can be helpful in some imbalanced settings but may also introduce limited gains or model-dependent trade-offs (Bajer et al., 2019; Chawla et al., 2002; Fernández et al., 2018b; Zhou et al., 2013). The present results therefore support a more cautious conclusion: resampling should be treated as a model-dependent design choice whose value must be verified empirically within a leakage-free pipeline rather than assumed a priori.

5.5. Theoretical implications

The revised findings have several implications for the theory and methodology of bankruptcy prediction under class imbalance.

First, the stability of model rankings across alternative cost structures indicates that cost-sensitive evaluation should be treated not merely as a fixed-parameter adjustment, but as a robustness problem in which conclusions should be tested under multiple plausible decision settings (Peykani et al., 2025; Radovanovic & Haas, 2023; Sahin et al., 2013). In this sense, the contribution of the study lies not in proposing a new algorithm, but in providing a more rigorous way to evaluate models when decision consequences are asymmetric.

Second, the calibration analysis highlights a rare-event phenomenon that is often underappreciated in bankruptcy prediction: severe class imbalance affects not only classification performance, but also the distribution and practical usefulness of calibrated probabilities. The present results indicate that threshold optimization should be interpreted jointly with calibration behavior rather than as an isolated post-processing step.

Third, the revised comparison between resampled and non-resampled configurations supports the broader view that data-level imbalance correction is not universally beneficial. Its usefulness depends on how it interacts with the classifier, the cost structure, and the evaluation criterion. This is theoretically important because it challenges the tendency to treat resampling as a default solution to class imbalance.

Fourth, the stronger and more stable performance of Random Forest relative to Logistic Regression is consistent with the broader literature showing that ensemble methods can better represent non-linear and interaction-based relationships in structured financial data (Barboza et al., 2017; Breiman, 2001). In bankruptcy prediction, such flexibility is especially relevant because distress often emerges from combinations of leverage, liquidity, profitability, and efficiency indicators rather than from simple linear effects alone (Sun et al., 2014).

Taken together, these findings suggest that bankruptcy prediction under extreme class imbalance should be conceptualized as a joint problem of discrimination, calibration, and decision robustness. This perspective extends beyond the use of any single metric and provides a more coherent theoretical basis for evaluating machine learning models in financially consequential settings.

5.6. Practical, policy, and ethical implications

From a practical perspective, the revised results reinforce the idea that bankruptcy prediction should be evaluated as a decision problem rather than solely as a classification problem. The most informative outcome is not simply that one model attains higher discrimination metrics than another, but that the relative ordering of models remains stable when misclassification costs are varied across a broad range. This stability matters because financial institutions often operate under uncertain or changing cost assumptions.

The results also clarify that discrimination, calibration, and cost-sensitive decision-making should be considered jointly. A model with stronger discrimination but weaker cost performance would not necessarily be preferable in operational settings. Conversely, a model with acceptable discrimination, lower expected cost, and more stable thresholds may provide a more reliable basis for deployment. In this sense, the revised framework supports a more decision-aware validation strategy that is consistent with recent calls for socially and economically grounded model evaluation in business failure prediction (Peykani et al., 2025; Radovanovic & Haas, 2023).

For financial institutions, the findings suggest several concrete recommendations. Model selection should be based on decision-oriented validation rather than on discrimination metrics alone. Candidate models should be evaluated using at least three groups of criteria: discrimination performance, calibration quality, and cost-sensitive decision performance. Cost ratios should be treated as operational scenario parameters rather than fixed constants, and threshold setting should be tested under multiple plausible cost scenarios before deployment. Calibrated probabilities should be monitored explicitly in operational use, and resampling should not be adopted as a default preprocessing rule but validated on a model-specific basis.

For regulators and supervisory bodies, the findings support validation frameworks that extend beyond accuracy-based reporting and include calibration diagnostics, threshold documentation, cost-sensitive evaluation, and preprocessing transparency. Institutions should be required to justify the rationale for the selected operating threshold and to demonstrate robustness under alternative cost assumptions. Explainability should be incorporated as a governance and audit-support tool, while model approval should still depend primarily on robust predictive and decision-oriented evidence.

The revised results also permit a scenario-based interpretation of expected cost. Under the primary 1:10 cost configuration, RF_balanced__none reduces expected cost by approximately 103–108 decision-weighted cost units relative to LR_balanced__none, depending on the correlation threshold. This corresponds to a relative reduction of roughly one-third. Although these values should not be interpreted as institution-specific monetary losses, they indicate that the reduction in expected cost is not merely statistically detectable but can also be operationally meaningful under plausible screening assumptions.

These implications also have an ethical dimension. Automated bankruptcy prediction systems may influence credit screening, monitoring intensity, escalation decisions, or perceptions of firm risk. When

such outputs are used in a highly automated manner, errors may have consequences that extend beyond model performance metrics, including restricted access to financing, premature risk escalation, or reduced managerial discretion in borderline cases. Because calibrated probabilities remain compressed and thresholds depend on explicit cost assumptions, model outputs should not be interpreted as neutral or context-free indicators of firm quality. Rather, they should be governed as transparent decision-support tools subject to documentation, review, and human oversight.

5.7. Limitations and future research

Several limitations should be acknowledged. First, the revised experiments are based on a single benchmark dataset. Although the Taiwanese Bankruptcy Prediction dataset remains useful for controlled methodological comparison, it reflects a specific institutional and temporal context, namely publicly listed firms in Taiwan over the period 1999–2009. This limitation is central to interpreting the scope of the present contribution. The findings should therefore be understood as evidence of methodological robustness within a benchmark setting rather than as a definitive statement on external generalizability across more recent market conditions, alternative institutional environments, or different regulatory regimes.

Second, the dataset does not provide institution-specific information on exposure, collateral, recovery, or realized loss. Accordingly, the cost ratios used in the revised analysis are best interpreted as stylized decision parameters rather than direct monetary loss estimates. Third, although the current revision substantially strengthens the analysis of calibration, cost sensitivity, and robustness, explainability is not yet incorporated as a formal criterion for model ranking.

The revised experiments improve internal robustness through repeated stratified cross-validation and sensitivity analyses across alternative cost ratios and correlation thresholds. However, these checks remain internal to the benchmark setting and do not substitute for external validation. Future research should therefore extend the framework to more recent and institutionally diverse datasets, including multi-country settings and alternative market environments. More specifically, external validation could proceed in at least three directions: first, testing the framework on newer bankruptcy datasets within the same institutional context to assess temporal stability; second, evaluating the framework on datasets from different countries or regulatory regimes to examine cross-context robustness under alternative accounting standards, legal environments, and bankruptcy definitions; and third, extending the analysis to other firm populations, such as SMEs or non-listed firms, where reporting quality, class prevalence, and distress dynamics may differ from those observed in the current benchmark. Further work may also investigate whether the probability compression and threshold behavior observed here persist in panel or time-series bankruptcy data, whether institution-specific cost structures alter the ranking of model configurations, and how more advanced explainability methods, including instance-level and counterfactual analysis, may further strengthen the transparency and practical usefulness of bankruptcy prediction systems.

6. Conclusion and future research

This study developed a leakage-free and cost-sensitive machine learning framework for corporate bankruptcy prediction under extreme class imbalance. By integrating feature de-redundancy, repeated stratified cross-validation, cost-sensitive learning, calibration analysis, and threshold optimization within a unified evaluation pipeline, the study addressed several methodological weaknesses that continue to affect bankruptcy prediction research, including data leakage, overreliance on conventional performance metrics, and insufficient attention to the decision consequences of misclassification.

The revised evidence shows that model assessment in this setting cannot rely on discrimination metrics alone. Across the benchmark experiments, the Random Forest configurations consistently delivered more favorable decision-oriented performance than the Logistic Regression baseline, particularly in terms of expected cost, minority-class detection, and threshold stability. The sensitivity analysis further showed that, although expected cost increases as the penalty for false negatives becomes larger, the relative ranking of the evaluated configurations remains substantively stable across alternative cost scenarios. These findings suggest that bankruptcy prediction models should be evaluated as decision-support tools

whose usefulness depends not only on predictive quality, but also on robustness under alternative operating assumptions.

The results also indicate that preprocessing choices should not be treated as universally beneficial by default. In the revised experiments, hybrid resampling did not consistently improve decision-oriented outcomes and did not alter the preferred model ranking. At the same time, the calibration analysis showed that predicted probabilities remain compressed under rare-event conditions, reinforcing the need for careful threshold selection, routine calibration monitoring, and explicit documentation of operating rules when such systems are used in practice.

From a practical perspective, the proposed framework offers actionable guidance for financial institutions seeking to implement data-driven early-warning systems. Cost-sensitive evaluation helps align predictive outputs with the asymmetric consequences of missed distress signals and false alarms, while explainability supports transparency, auditability, and managerial trust. For regulators and supervisory bodies, the findings support validation frameworks that extend beyond discrimination metrics to include calibration diagnostics, threshold justification, preprocessing documentation, and robustness under multiple cost assumptions. More broadly, the study suggests that automated bankruptcy prediction should be governed as a transparent decision-support mechanism, since risk scores, calibration choices, and threshold settings may carry broader ethical and societal consequences in high-stakes financial contexts.

Several limitations remain. The empirical evidence is derived from a single benchmark dataset covering the period 1999–2009 and should therefore be interpreted as within-benchmark evidence of methodological robustness rather than as a final statement on external generalizability across countries, industries, or regulatory environments. In addition, the cost ratios used in this study are best understood as stylized decision parameters rather than institution-specific monetary loss estimates. The most important next step for future research is external validation on more recent and institutionally diverse datasets. Further work should also evaluate alternative random-seed settings and institution-specific cost structures, explore dynamic extensions based on panel or time-series data, and examine more advanced explainability approaches, including instance-level and counterfactual analysis, to strengthen the transparency and practical usefulness of bankruptcy prediction systems.

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T.T.D.: Conceptualization, Data curation, Formal analysis, investigation, Writing – original draft; X.T.D.: Project administration, Supervision, Writing – original draft, Writing – review & editing.

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Data availability statement

The original dataset used in this study is the Taiwanese Bankruptcy Prediction dataset, publicly available from the UCI Machine Learning Repository (https://doi.org/10.24432/C5004D). The data-processing scripts, analysis code, and derived outputs supporting the findings of this study are available from the corresponding author upon reasonable request.

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